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📁 WORK EXPERIENCE

Senior Implementation Tech Lead — Can/Am Technologies — Highlands Ranch, CO *April 2025 - Present*

- Leading a team of forward-deployed engineers responsible for integrating with third-party systems.
- Formalising a per-project *Integrations Lead* role definition for the person responsible for overseeing technical delivery of the business solution to the client.
- Encouraging adoption of AI efficiencies across the Implementations engineering team.
- Collaborating with cross-functional stakeholders to improve processes and standards across the client services organisation.
- Driving multiple efforts to make integration architecture more declarative and configuration-driven enabling its amenability to AI-assisted programming and configuration.
- Implementing improved developer tooling such as automated linting for integration development.

Software Engineer — Beeper — Remote *July 2021 - April 2025*

- Scaled our backend infrastructure from handling <1,000 users to >100,000 users by **sharding** traffic from high-volume bridges to a separate **Go** service called *Hungryserv* in a backwards-compatible, transparent manner. I created the initial proof of concept and then continued as a core member of the 3-member team that productionised the project over a four-month period.
- Reduced RAM usage for the **Telegram** to **Matrix** bridge by ~2TB (80%) by rewriting from Python to **Go**.
- **Reverse-engineered** and implemented features for *Beeper Mini* (iMessage on Android) including media, tapbacks, typing indicators, read receipts, edits, unsends, link previews, and chat metadata changes.
- Implemented the cryptographic key infrastructure necessary for message key backups and interactive device verification in *mautrix-go* by utilising the standard **Go cryptography libraries**.
- Implemented media upload/download and interactive device verification in the Beeper client SDK written in **Go** which is being used in the next generation Beeper clients.
- Measured message send **latency and reliability** by instrumenting bridge **metrics**. Built a Dockerised **Go** service to process those metrics and send them to BigQuery.
- **Reverse-engineered** the LinkedIn Messaging API and implemented a LinkedIn to Matrix bridge in **Python**.
- Designed a framework for importing users' chat history, and implemented it in multiple bridges.

Adjunct Professor — Colorado School of Mines — Golden, CO *August 2018 - December 2024*

- **Algorithms** (4×) — advanced data structures, graph algorithms, dynamic programming, NP-completeness
- **Programming Languages** (4×) — functional programming, parsers, type systems, formal semantics
- **Computer Organization** (1×) — RISC-V assembly, pipelining, processor design, memory hierarchy
- **Computer Architecture** (1×) — cache coherence, virtual memory, branch prediction, multiprocessors

Software Engineer — The Trade Desk — Denver, CO *June 2019 - July 2021*

🎓 EDUCATION

Colorado School of Mines — B.S. + M.S. in Computer Science — Golden, CO *July 2016 - May 2019*

📄 TALKS & PRESENTATIONS

Matrix Cryptographic Key Infrastructure — Matrix Conference *September 2024*

- Provided an overview of how key sharing, key backup, device and user verification, and secret storage operate within Matrix to provide cryptographically secure messaging features.

Hungryserv: A Homeserver Optimized for Unfederated Use-Cases — Berlin Matrix Summit *August 2022*

- Discussed a Matrix-compatible homeserver that Beeper uses to handle unfederated bridge traffic.